

Prepare and Plate a Sandwich SOP

One episode builds one sandwich and puts it on a plate. Two slices of bread, one spread, and one slice of ham come off the ingredient tray in a fixed order, get closed, cut once on the diagonal, and end up as two halves on the serving plate. The scraps are swept into one pile on the left of the board and the knife is back in its spot when the episode ends.

The build never changes. Bottom slice, spread, ham, top slice, cut, plate.

The knife does three jobs: it spreads, it cuts, and it sweeps the scraps aside. It is held two ways for them, loose at the butt of the handle for spreading and close to the blade for cutting and sweeping, and it returns to its spot between jobs. No grasp is ever changed while something is in the air.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the ingredient tray on the left, the knife spot, the cutting board in the middle, and the serving plate on the right.
3. Both arms are at home with grippers open.
4. The tabletop is clear of anything but the zones listed below.
5. Nothing obstructs the paths between the tray, the board, and the plate.

Materials checklist

1. The ingredient tray sits against the left edge and holds three items, all lying flat and none overlapping: two slices of bread in the back row and one slice of ham in the front row. The **staging spot**, the clear patch at the tray's outer end, is empty. It is the only place on the tray a gripper may release, and it is used only to lift a covering item clear.
2. The two bread slices are the same size and are not torn. The slice nearer the board is the bottom slice for this episode.
3. The spread tub sits open on the inner edge of the tray, the edge nearer the board, within reach of both arms. The lid is off and set aside off the tray. The tub holds enough spread to cover a whole slice with room to spare, so nothing has to be scraped out.
4. The knife lies flat in its knife spot, on the clear patch between the tray and the board, with the whole handle clear of anything else so a gripper can take it from above. There is no rest, cradle, or slot: the knife is simply set down. Its handle is bulky enough to grasp off a flat surface. The blade is clean and dry.

5. The cutting board lies flat in the middle of the tabletop and is bare.
6. The serving plate sits against the right edge, empty and right side up.

Workspace layout

- **Ingredient tray:** on the tabletop against the left edge, within reach of the left arm. It holds the two bread slices, the ham, the open spread tub, and the staging spot. The tub sits on the tray's inner edge so the right arm can reach it to hold it down. The staging spot is the clear patch at the tray's outer end, kept empty for overlap recovery.
- **Knife spot:** the clear patch of tabletop between the tray and the board, within reach of the left arm. The knife lies flat here whenever it is not in a gripper. It is a bare patch, not a rest or a holder.
- **Cutting board:** flat in the middle of the tabletop, reachable by both arms. The whole build happens here. It is bare at the start, and at the end it carries one pile of scraps on its left side and nothing else.
- **Serving plate:** on the tabletop against the right edge, within reach of the right arm.

Ownership rule, stated once. The left arm owns the tray and the knife. The right arm owns the plate. The board is shared and the spread tub is shared, and both arms may enter a shared zone, but only to do the actions assigned to them there. Neither arm takes over the other's work in a shared zone.

Vocabulary

Arm assignments

- **The left gripper owns the tray and the knife.** It brings every ingredient onto the board, does every spread, makes the cut, and sweeps the scraps aside. It never reaches the plate.
- **The right gripper owns the plate.** It holds the sandwich still during the cut and carries both halves to the plate. It never picks anything off the tray and never touches the knife.
- **The board and the spread tub are shared.** Both arms work over the board, each doing only its own assigned actions there. The right gripper's one job past the board is holding the spread tub down while the knife loads, and it holds the tub without ever taking hold of anything on the tray.
- **Nothing is passed hand to hand.** Every item that changes arms does so by sitting on the board.
- A gripper releases only onto the board, onto the plate, onto the knife spot, or into the tray staging spot during an overlap recovery. Never open a gripper over open space, over the rest of the tray, or over the spread tub.

Handling the knife

The knife does three jobs and is held two ways. The two grips are not interchangeable.

- **The knife is taken by the handle, off a flat surface, every time.** It lies flat in its knife spot and the gripper comes straight down on the handle. The handle is bulky enough that this needs no rest, cradle, or slot.
- **Spread grip:** the left gripper takes the handle at its **butt end**, the end furthest from the blade. This is the grip for loading spread and drawing it across the bread.
- **Cut grip:** the left gripper takes the handle **just behind the blade**. This is the grip for the press cut and for the sweep, and it is the only grip that gives a steady press.
- **Change grips on the knife spot, never in the air.** Lay the knife back down flat and release before taking the other grip.
- **Two blade poses, and the wrist is what changes them.** The knife comes up **flat**, the way it was lying, and stays flat for loading and spreading. For the cut and for the sweep the wrist turns it **edge down** on the way over. Turn the wrist while carrying, clear of the board, and never shift the knife inside the fingers to do it.
- The knife is never carried over the plate, never left lying on the cutting board, and never left in a gripper while an ingredient is being moved.

The build order

- **Layer order, not reach order.** Layers land in this order and no other: bottom slice, spread, ham, top slice. This rule governs the order things are **placed onto the stack**, not the order they are picked up. The top slice is fetched early and staged on the board in Step 1, and it still goes onto the stack last.
- The **slice nearer the board is the bottom slice** and the slice further from the board is the top slice. That is fixed at setup and does not change mid episode.
- Never take two items off the tray in one trip, and never lift a slice that has another item resting on it. If items have shifted into a stack on the tray, lift the top one clear into the **tray staging spot** and release it there, then take the item that was due. This is the only release permitted over the tray.
- Every layer lands **centered and square on the layer below it**, with the bread edges roughly in line rather than fanned out.

Handling standard

- Move exactly one item at a time.
- Grip the bread and the ham **by an edge**, close enough to the rim that the gripper is not pressing on the middle of the slice.
- Carry every item level and lower it straight down. Lower each item until it is resting on the layer below, then release. Never drop an item from above.
- Nothing is slid, and the only two sliding motions in the whole episode are made by the knife: drawn across the bread during a spread, and drawn across the board during the sweep. Correct a bad placement by lifting the item clear and setting it down again.

- **Do not change the grasp while an item is in the air.** Taking a fresh grasp after an item is resting is permitted, and is how every bad placement is corrected.
- A layer is settled when it stays still for 2 seconds after release.
- **Never press, flatten, or straighten a layer with the gripper.** The bread is soft enough that a gripper press dents or tears it. The right gripper holding the sandwich still for the cut is the only time a gripper is permitted to bear on the assembled sandwich at all. Knife contact is a separate thing and is governed by Steps 2 and 5.

Steps

Step 1: Lay the two bread slices

Goal: the bottom slice is centered on the board and the top slice is staged behind it.

- The **left gripper** grips the bottom slice by its edge, lifts it straight up off the tray, and carries it level to the board.
- Lower it flat and centered on the board, then release.
- The **left gripper** grips the top slice by its edge, lifts it straight up, and lays it flat on the back half of the board, behind the bottom slice and not touching it.

The top slice is only **staged** here. It is fetched now so the tray is cleared in one visit, and it goes onto the stack last, in Step 4. Nothing is spread on it and nothing is laid on it.

Check: both slices lie flat and separate, the bottom slice is centered on the board, and neither slice hangs over an edge. If a slice lands folded or overhanging, lift it clear and lay it down again.

Step 2: Spread the bottom slice

Goal: the whole face of the bottom slice is covered with spread and the knife is back in its spot.

2.1 Load the knife

- The **left gripper** takes the knife in the **spread grip**, at the butt of the handle, and lifts it straight up off its knife spot.
- The **right gripper**, closed and empty, comes down on the rim of the spread tub and holds the tub down against the tray. It takes this hold **before the blade goes in**.
- The **left gripper**, keeping the blade **flat** as it was picked up, lowers it into the tub, presses it into the spread, and lifts it straight out.
- The **right gripper** lifts clear of the tub only once the blade is out and clear of the rim.

The tub is light and the blade drags on the way out, so an unheld tub lifts with the knife or slides off the tray. Anchor it every time, even when the spread is soft, and on every reload.

2.2 Draw the spread across

- Carry the knife level to the board, blade still flat, and lay it on the near edge of the bottom slice so the **flat of the blade** meets the bread.
- Draw the blade across the face of the slice, from the near edge to the far edge, keeping it flat.
- Work across the slice in passes until the whole face is covered, corner to corner and out to the crust, with no bare bread showing.
- Lift the blade clear of the slice before moving away.

Full coverage is the standard, not a pass count. Take as many passes as the slice needs.

If the knife runs dry before the face is covered, reload it through Step 2.1, tub hold included, and carry on where you stopped. It stays in the spread grip between loads.

Keep the spread on the bread. Do not run the blade off the crust onto the board.

2.3 Return the knife

- Carry the knife back to its knife spot, lay it down flat, and release.

Check: the face of the bottom slice is covered out to the crust with no bare patches, the knife is lying flat in its knife spot, the spread tub is still upright on the tray's inner edge, the bottom slice is still centered and flat on the board, and no spread has been dragged onto the board. If a bare patch is left, take the knife back up and cover it before moving on. If the slice was pushed out of place by the spreading, lift it and center it again before continuing. If the tub has shifted, set it back on the tray's inner edge before continuing.

Step 3: Layer the ham

Goal: the ham sits centered on the spread face of the bottom slice.

- The **left gripper** grips the ham by its edge, lifts it straight up off the tray, and carries it level to the board.
- Lower it centered and flat on the spread face, then release once it is resting.
- Leave the ham where it lands. Do not nudge, flatten, or straighten it.

Check: the ham sits on the spread roughly square with the bread rather than turned across it, and it does not overhang the bread far enough to fall. If it lands folded over on itself or hanging off the slice, lift it clear by its edge and lower it again.

Expected state: the tray holds nothing but the open spread tub. The board holds the loaded bottom slice in the middle and the plain top slice behind it. The knife is lying in its knife spot.

Step 4: Close the sandwich

Goal: the top slice is square on the stack and the sandwich is closed.

- The **left gripper** grips the top slice by its edge, lifts it straight up off the back of the board, and carries it level over the stack.
- Lower it flat onto the ham so its edges line up with the bottom slice, then release.
- Lift the gripper clear. The sandwich is closed by the weight of the top slice alone. Nothing presses it down.

Check: the sandwich is one square stack with the two bread edges in line, the ham is not fanned out past the bread far enough to hang loose, and the stack sits flat rather than sliding apart. If the top slice is clearly off the stack, lift it and lay it down again. Do not shove it into line.

Step 5: Cut it diagonally

Goal: one cut line corner to corner, leaving two halves on the board.

5.1 Take the cut grip

- The **left gripper** takes the knife in the **cut grip**, on the handle just behind the blade, and lifts it straight up off its knife spot.
- Turn the wrist as the knife comes up so the blade hangs **edge down**. Do this on the way, clear of the board, and never by shifting the knife inside the fingers.

5.2 Hold and cut

- The **right gripper**, closed and empty, rests on the half of the sandwich nearer the plate and holds it still for the whole cut. Rest it with just enough weight to stop the stack sliding. Do not bear down on it.
- The **left gripper** brings the blade over the sandwich, edge down, and lines it up **corner to corner**, from the near corner to the far corner across the diagonal.
- Press the blade straight down through the sandwich until it meets the board. Do not saw the blade back and forth.
- Lift the blade straight up and clear, then the **right gripper** lifts clear.

One cut line, up to three presses. Make one press. If the halves are still joined, press straight down again through the **same line**, up to three presses in total. Do not start a second line and do not pull the halves apart by hand. If the halves are still joined after the third press, the blade is too dull: stop, end the episode, and log it as a knife fault.

5.3 Return the knife

- The **left gripper** carries the knife back to its knife spot, lays it down flat, and releases.

Check: the sandwich is in two halves, cut along one line from corner to corner, and the knife is lying flat in its knife spot. Both halves are still on the board.

Step 6: Plate the two halves

Goal: both halves lie flat on the plate, cut edges facing each other.

Work one half at a time, taking the **half nearer the plate first**.

- The **right gripper** grips the half by the crust on its long outer edge, lifts it straight up off the board, and carries it level to the plate.
- Lower it flat on the plate and release. The first half goes on the near side of the plate; the second half goes beside it with its **cut edge facing the first half's cut edge** and its point aimed the same way.
- Carry each half level so the layers stay stacked. Do not tilt it up on its cut edge.

Check: both halves lie flat on the plate, cut edges facing each other, points aimed the same way, and neither half overhangs the rim. Nothing has slid out from between the bread. If a half lands turned the wrong way or off the plate, lift it clear and lower it again.

Expected state: the plate holds both halves. The board holds nothing but loose scraps. The knife is in its knife spot and the tray holds nothing but the open spread tub.

Step 7: Sweep the scraps to one side of the board

Goal: every scrap is gathered into one pile against the left side of the board, the rest of the board is clear, and both arms are safely home.

The knife is the sweeping tool. Nothing is picked up by hand in this step, whatever its size. Nothing leaves the board either: the scraps stay on it, gathered to one side, and they are cleared off during the reset.

7.1 Sweep

- The **left gripper** takes the knife in the **cut grip**, on the handle just behind the blade, and lifts it straight up off its knife spot.
- Turn the wrist so the blade is **edge down**, then set it on the board at the **right edge**, angled across the board.
- Draw the blade **right to left** across the board, gathering everything in its path ahead of it. Keep the blade in contact with the board for the whole stroke.
- End each stroke **just inside the left edge**, not over it. The pile stays on the board.
- Repeat until the rest of the board is clear and everything sits in one pile on the left. Sweep the far half and the near half on separate strokes rather than trying to take the whole depth at once.
- Sweep fallen ham and torn bread the same way as crumbs. Do not stop to pick anything up.

7.2 Put the knife away

- Carry the knife back to its knife spot, lay it down flat, and release.

Check: the scraps sit in one pile against the left side of the board, the rest of the board is clear, nothing was pushed off any edge onto the tabletop, and the knife is lying flat in its knife spot.

7.3 End the episode

- Confirm the board carries one pile on the left and nothing anywhere else on it, and that nothing was left on the tray but the spread tub.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Clear the plate and set the two halves aside for disposal.
2. Clear the pile of scraps off the board and into the bin.
3. Wipe the board clean and dry, and put it back flat in the middle of the tabletop.
4. Wipe the knife clean and dry and lay it flat on its knife spot.
5. Restock the tray: two fresh bread slices flat in the back row and one ham slice flat in the front row, none of them overlapping, and the staging spot at the tray's outer end left clear. Vary how the slices sit between episodes rather than rebuilding the same layout every time.
6. Top up the spread tub if it no longer holds enough to cover a whole slice, and leave it open on the tray with the lid off the tray.
7. Replace any slice that is torn, dried out, or stuck to another slice.
8. Wipe the tabletop and put the plate back empty against the right edge.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Built out of order

- **Visible cue:** the layers go on in any order other than bottom slice, spread, ham, top slice, such as the ham before the spread, or the top slice fetched early and set on the stack.

- **SOP rule broken:** Steps 1 to 4, layer order, not reach order: bottom slice, spread, ham, top slice.
- **Coaching note:** name the next layer before you reach for it.

Violation: Wrong slice used as the bottom

- **Visible cue:** the slice further from the board is taken first and spread, or the two slices are swapped part way through the build.
- **SOP rule broken:** Step 1, the slice nearer the board is the bottom slice, and that is fixed at setup.
- **Coaching note:** the near slice is the base, every episode.

Violation: Arm crossed into the other arm's side

- **Visible cue:** the left gripper reaches the plate or places a half on it; or the right gripper picks something off the tray, takes the knife, or makes the sweep. Holding the spread tub down during Step 2.1 is not this violation.
- **SOP rule broken:** Steps 1 to 7, the left arm owns the tray and the knife, the right arm owns the plate, and either arm enters the board or the spread tub only to do the actions assigned to it there.
- **Coaching note:** left arm builds, right arm plates. Sharing the board is not taking over the work done on it.

Violation: Spread tub not held during loading

- **Visible cue:** the blade goes into the tub with no right gripper on it, the right gripper arrives after the blade is already in, or it lifts away while the blade is still in the spread. The tub lifts with the knife, slides, or tips.
- **SOP rule broken:** Step 2.1 (the right gripper holds the tub down against the tray before the blade goes in and lifts clear only once the blade is out and clear of the rim, on every load including reloads).
- **Coaching note:** anchor the tub first, release it last.

Violation: Spread tub gripped or moved

- **Visible cue:** the right gripper closes on the tub and picks it up, drags it across the tray, repositions it, or holds it somewhere other than down against the tray.
- **SOP rule broken:** Step 2.1, the right gripper holds the tub down without ever taking hold of anything on the tray.
- **Coaching note:** hold it down, do not pick it up.

Violation: Item passed hand to hand

- **Visible cue:** an ingredient, a sandwich half, or the knife is transferred directly from one gripper to the other in mid air instead of being set down on the board first.

- **SOP rule broken:** Steps 1 to 6, nothing is passed hand to hand, and every item that changes arms does so by sitting on the board.
- **Coaching note:** the board is the handoff. Nothing changes hands in the air.

Violation: Wrong knife grip for the job

- **Visible cue:** the knife is spread with while held on the handle just behind the blade, or the cut or the sweep is made while held at the butt of the handle. The gripper closing anywhere on the blade rather than on the handle is this violation too.
- **SOP rule broken:** Steps 2.1, 5.1, and 7.1, spread grip at the butt of the handle, cut grip just behind the blade for both the cut and the sweep.
- **Coaching note:** butt end to spread, close to the blade to cut.

Violation: Knife regripped in the air or left off its spot

- **Visible cue:** the gripper shifts its hold along the handle without setting the knife down, the knife is worked around inside the fingers to change the blade pose, the knife is parked on the cutting board or the tray between uses, or it is left in a gripper while an ingredient is moved.
- **SOP rule broken:** Steps 2.3 and 5.3, change grips on the knife spot and never in the air, and the knife lies flat in its knife spot whenever it is not in use.
- **Coaching note:** the knife goes home between every job.

Violation: Wrong blade pose for the job

- **Visible cue:** the spread is drawn with the blade edge down instead of flat, or the cut or the sweep is made with the blade still flat instead of turned edge down. The blade is also turned while it is already touching the bread or the board rather than on the way over.
- **SOP rule broken:** Steps 2.2, 5.1, and 7.1, flat for loading and spreading, and the wrist turned edge down on the way over for the cut and the sweep, turned while carrying and clear of the board.
- **Coaching note:** flat to spread, turned edge down to cut and sweep, and turn it before you arrive.

Violation: Bottom slice left partly bare

- **Visible cue:** the knife goes back to its spot with bare bread still showing on the face, spread stopping short of the crust on one or more edges, or a gap between passes left uncovered. The ham then goes down on a partly bare slice.
- **SOP rule broken:** Step 2.2 (work across the slice in passes until the whole face is covered, corner to corner and out to the crust, with no bare bread showing).
- **Coaching note:** check the whole face before you put the knife down.

Violation: Reload skipped or done wrong

- **Visible cue:** a gripper keeps dragging an empty blade across the slice instead of going back to the tub, scrapes the blade on the crust to work the last of the spread off it, or reloads without the right gripper holding the tub, or parks the knife on its spot between loads.
- **SOP rule broken:** Step 2.2 and Step 2.1 (if the knife runs out before the face is covered, carry it back to the tub and load it again following all of Step 2.1 including the right gripper hold; the knife stays in the spread grip and does not go back to its spot between loads).
- **Coaching note:** empty blade means go back to the tub, not press harder.

Violation: Spread run off the bread

- **Visible cue:** the blade runs off the crust and smears spread onto the board, or spread is worked over the edge and down the side of the slice.
- **SOP rule broken:** Step 2.2 (keep the spread on the bread; do not run the blade off the crust onto the board).
- **Coaching note:** stop the stroke at the crust.

Violation: Layer adjusted after release

- **Visible cue:** the gripper nudges, flattens, straightens, or pats the ham or a bread slice after releasing it, or presses down on the closed sandwich to flatten it, instead of leaving it where it landed or lifting it clear and lowering it again.
- **SOP rule broken:** Steps 3 and 4, leave the ham where it lands, correct a bad placement by lifting the item clear and setting it down again, and never press or flatten a layer with the gripper.
- **Coaching note:** lift it and place it again, or leave it alone. Bread dents and does not come back.

Violation: Sandwich not held during the cut

- **Visible cue:** the blade comes down on a sandwich no gripper is holding, and the stack slides, spins, or splays apart under the press.
- **SOP rule broken:** Step 5.2 (the right gripper rests on the half nearer the plate and holds it still for the whole cut, with just enough weight to stop the stack sliding).
- **Coaching note:** the right hand anchors the sandwich before the blade touches it.

Violation: Cut is not a single diagonal line

- **Visible cue:** the blade is sawed back and forth, the cut runs straight across instead of corner to corner, a second cut line is started, the halves are torn apart by hand after a partial cut, or a fourth press is made on the same line instead of stopping.
- **SOP rule broken:** Step 5.2 (line the blade up corner to corner and press straight down through to the board; one cut line, up to three presses on that same line; stop and log a knife fault if the halves are still joined after the third).

- **Coaching note:** one line, straight down, all the way to the board.

Violation: Halves plated wrong

- **Visible cue:** a half is laid on the plate with its cut edge facing out, the two points aimed different ways, the halves stacked on each other, a half tipped up on its cut edge, or a half overhanging the rim.
- **SOP rule broken:** Step 6 (both halves flat on the plate, cut edges facing each other, points aimed the same way, nothing overhanging the rim; carry each half level).
- **Coaching note:** cut edges face each other, points the same way, both flat.

Violation: Wrong plating order

- **Visible cue:** the half further from the plate is carried over first, or both halves are lifted together.
- **SOP rule broken:** Step 6, take the half nearer the plate first and move exactly one item at a time.
- **Coaching note:** near half first, one trip each.

Violation: More than one item moved at once

- **Visible cue:** two ingredients are lifted off the tray together, both sandwich halves are carried in one trip, or an ingredient is moved while the knife is still in the other gripper.
- **SOP rule broken:** Steps 1 to 6, move exactly one item at a time, and the knife is never held while an ingredient is moved.
- **Coaching note:** one item per trip, every trip.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the board, the stack, or the plate and the item falls the rest of the way, instead of being lowered until it is resting.
- **SOP rule broken:** Steps 1 to 6, lower each item until it is resting on the layer below, then release, and never drop an item from above.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, drags, or shoves a slice, the ham, or a half across the tray, the board, or the plate to line it up, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 1 to 6, nothing is slid except the knife during a spread, and correct a bad placement by lifting the item clear and setting it down again.
- **Coaching note:** lift it clear and put it down again. Nothing gets shoved into position.

Violation: Item taken from under another on the tray

- **Visible cue:** a slice is pulled out from under another item that has shifted on top of it, it is dragged sideways across the tray to free it, or the covering item is released somewhere on the tray other than the staging spot, instead of being lifted clear into the staging spot first.
- **SOP rule broken:** Steps 1 to 4, never lift a slice that has another item resting on it, lift the top one clear into the tray staging spot and release it there, then take the item that was due.
- **Coaching note:** uncover it, then lift it. Never tug it free.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the tray, over the spread tub, or over open tabletop, or an item is set down anywhere other than the board, the plate, or the knife spot.
- **SOP rule broken:** Steps 1 to 7, a gripper releases only onto the board, onto the plate, or onto the knife spot.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Board not swept clear

- **Visible cue:** the episode reaches the end with scraps still scattered across the board rather than gathered into one pile on the left, or the sweep stops while crumbs, fallen ham, or torn bread are still sitting out on the open board.
- **SOP rule broken:** Step 7.1 (repeat the stroke until the rest of the board is clear and everything sits in one pile on the left).
- **Coaching note:** keep sweeping until it is one pile, not until it looks tidy.

Violation: Scraps picked up instead of swept

- **Visible cue:** the gripper closes on a piece of ham, bread, or a crumb and carries it away, or a bare gripper is used to brush or wipe the board instead of the knife.
- **SOP rule broken:** Step 7 (the knife is the sweeping tool; nothing is picked up by hand in this step, whatever its size).
- **Coaching note:** the knife gathers the pile. Fingers never touch a scrap.

Violation: Scraps pushed off the board

- **Visible cue:** a stroke carries debris over the left edge, a side edge, or the far edge and puts it on the tabletop, into the ingredient tray, or onto the floor, instead of stopping just inside the left edge.
- **SOP rule broken:** Step 7.1 (end each stroke just inside the left edge, not over it; the pile stays on the board).
- **Coaching note:** stop the stroke short. The board keeps the scraps.

Violation: Swept in the wrong direction

- **Visible cue:** strokes run left to right, toward the plate, or toward the front edge, instead of right to left across the board, or the pile ends up somewhere other than the left side.
- **SOP rule broken:** Step 7.1 (set the blade at the right edge and draw it right to left, gathering everything ahead of it into one pile on the left).
- **Coaching note:** always right to left, always into the same corner.

Violation: Blade lifted off the board mid sweep

- **Visible cue:** the blade skips or hops during a stroke, riding over debris and leaving it behind, instead of staying in contact with the board for the whole stroke.
- **SOP rule broken:** Step 7.1 (keep the blade in contact with the board for the whole stroke).
- **Coaching note:** stay down on the board from the right edge to the end of the stroke.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item falls off the board or the plate, the knife strikes the board edge or the plate, the spread tub is knocked over, a layer is swept off the stack, or a sandwich half comes apart in transit.
- **SOP rule broken:** Steps 1 to 7, keep each item controlled on a clear travel path and release only after a stable placement, and the knife is never carried over the plate.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the knife left off its knife spot, a half still on the board, scraps still scattered outside the pile, an arm away from home.
- **SOP rule broken:** Step 7.3 (confirm the knife on its knife spot, one pile on the left of the board and nothing else on it, and the plated halves; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective ingredient:** a slice found torn, dried hard, or stuck to another slice mid episode, or ham that comes off the tray in pieces. Replace it before the next episode.
- **Spread tub runs empty** before the face of the slice is covered, leaving nothing left to load the knife with. Top it up before the next episode.

- **Knife fault:** a blade that has not parted the halves after three presses on the same line, or a handle too slick or too slim for the gripper to lift off a flat surface. Sharpen or replace the knife before the next episode.
- **Board or plate shifts out of its zone** during the episode.

Annotation subtasks (from SOP)

1. Lift a covering item clear on the ingredient tray
2. Move one bread slice from the tray to the board
3. Hold the spread tub down while the knife loads
4. Take the knife off its spot and load it with spread
5. Draw the spread across the bottom slice
6. Return the knife to its spot
7. Move the ham from the tray to the bottom slice
8. Close the sandwich with the top slice
9. Hold the sandwich still for the cut
10. Press cut the sandwich corner to corner
11. Move one sandwich half from the board to the plate
12. Sweep the scraps into one pile on the left of the board with the knife
13. Return both arms home and end the episode